

2025

Denmark: Roskilde Bank Emergency Liquidity Program, 2008

Bailey Decker

YPFS, Yale School of Management

Follow this and additional works at: <https://elischolar.library.yale.edu/journal-of-financial-crises>



Part of the [Banking and Finance Law Commons](#), [Economic History Commons](#), [Finance Commons](#), [Finance and Financial Management Commons](#), [Law and Economics Commons](#), [Policy Design, Analysis, and Evaluation Commons](#), and the [Public Administration Commons](#)

Recommended Citation

Decker, Bailey (2025) "Denmark: Roskilde Bank Emergency Liquidity Program, 2008," *Journal of Financial Crises*: Vol. 7 : Iss. 1, 166-188.

DOI: <https://doi.org/10.17132/2693-3179.1617>

Available at: <https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/7>

This Case Study is brought to you for free and open access by the Journal of Financial Crises and EliScholar – A Digital Platform for Scholarly Publishing at Yale. For more information, please contact journalfinancialcrises@yale.edu.

READERS TAKE NOTICE:

As of April 2024, the YPFS Resource Library's site domain has changed to <https://elischolar.library.yale.edu/ypfs-financial-crisis-resource-library/>.

Please be aware that upon clicking any of the URL references to the former Resource Library domains, either <https://ypfs.som.yale.edu> or <https://ypfsresourcelibrary.blob.core.windows.net/>, in the "References/Key Program Documents" section of a case study, readers will encounter a "page not found" error.

Readers can still retrieve a given resource cited within this case study on the [new site](#) by searching [here](#) for the title cited.

Denmark: Roskilde Bank Emergency Liquidity Program, 2008¹

*Bailey Decker*²

Yale Program on Financial Stability Case Study
April 15, 2025

Abstract

Roskilde Bank A/S (Roskilde) was the eighth-largest bank in Denmark at the time of the Global Financial Crisis, with approximately 43 billion Danish kroner (DKK; USD 9.1 billion) in consolidated assets as of March 2008. Roskilde had considerable exposure to real estate and construction firms, prompting ratings downgrades and larger write-downs than expected in July 2008. On July 10, 2008, Roskilde asked for liquidity assistance from the Danish central bank, Danmarks Nationalbank (DNB). Later that day, DNB and the banking sector's self-insurance group, the Private Contingency Association (PCA), announced emergency liquidity assistance to Roskilde in the form of an unlimited credit facility, conditioned on Roskilde's full or partial sale. When a private buyer did not emerge, DNB and the PCA took over Roskilde on August 24, 2008, and created a bridge bank. Danish authorities transferred most of the assets and liabilities from Roskilde (Old Roskilde) into a new bridge bank, New Roskilde. The PCA converted its guarantee to an equity stake in the bridge bank. The outstanding DKK 14.5 billion on the credit facility was transferred to New Roskilde; DNB also used the credit facility to repay all the senior liabilities of New Roskilde. A new resolution authority, the Financial Stability Company (FSC), took over New Roskilde and the credit facility from DNB in August 2009. New Roskilde continued to borrow emergency liquidity from the FSC until the FSC merged New Roskilde's assets into a liquidation vehicle in 2011. DNB and the FSC ultimately converted DKK 8.7 billion of debt from the credit facility into equity capital. At the end of 2010, the Danish state's total investment in Roskilde Bank's equity and subordinated debt reached DKK 12.4 billion and the state's outstanding loss on the restructuring of Roskilde, from the credit facility and direct equity injections, stood at DKK 8.9 billion. It is unclear if asset sales allowed the FSC to recoup some losses or whether the state's remaining loans to the bank were fully repaid, because the FSC has consolidated the financial reporting for the multiple banks that it acquired.

Keywords: Denmark, emergency liquidity assistance, Private Contingency Association, resolution, Roskilde Bank

¹ This case study is part of a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering ad hoc emergency liquidity provision. A survey of all the cases in this series (Kelly et al. 2025) and the individual cases underlying it are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/>. Wiggins et al. (2022) surveys broad-based emergency liquidity programs.

² Research Associate, YPFS, Yale School of Management.

Overview

This module describes the ad hoc emergency liquidity (AHEL) provided to Roskilde Bank (Roskilde) in 2008. Roskilde was subsequently nationalized, partially sold, and resolved. For a case study dedicated to Roskilde's restructuring and resolution, see Decker (2024).

Roskilde was the eighth-largest bank in Denmark, with just less than DKK 43 billion (USD 9.1 billion)³ in consolidated assets as of March 2008 (NAO 2009).

Roskilde had considerable exposure to real estate and construction firms in the summer of 2008 (AFP 2008; DFSA 2009; Ministry of Business and Growth 2013; Sjögren and Iversen 2019). Real estate assets comprised more than 43% of Roskilde's loan and guarantee portfolio at the time, which was the main contributor to Roskilde's failure since it had almost no US subprime asset exposure and ample access to central bank funds (AFP 2008; FT 2008; Ministry of Business and Growth 2013). Throughout the mid-2000s, Roskilde was one of the two fastest-growing regional, mid-sized banks amid a nationwide boom in expanding credit (see Figure 1) (DFSA 2009; Ministry of Business and Growth 2013; Sjögren and Iversen 2019). The US housing market crisis and subsequent credit crunch caused Roskilde to take larger write-downs than expected (DNB 2009b; NAO 2009).

On July 1, 2008, Moody's Investors Service downgraded Roskilde citing the bank's real estate exposure and weakening asset quality (Moody's 2008). Roskilde later said that it had explored the possibility of raising additi-

Key Terms

Purpose: To provide the necessary liquidity for Roskilde's day-to-day activities; later, to redeem senior creditors

Announcement Date July 10, 2008

Operational Date July 10, 2008

Termination Date The credit facility was transferred to the FSC in August 2009 and remained available until New Roskilde's assets and liabilities were subsumed by FS Finans and FS Bank in 2011

Legal Authority DNB could grant advances against security, according to Section 22 of the central bank bylaws

Administrator DNB and the PCA

Peak Authorization Unlimited credit facility

Peak Outstanding DKK 36.8 billion

Collateral Roskilde did not pledge collateral; the liquidity facility was secured by the PCA and state guarantees

Haircut and Recourse Not applicable

Interest Rate and Fees 6.6% interest rate; no fees

Term Unlimited in duration; still outstanding in 2011

Part of a Package State guarantee of credit facility; deposit guarantee; sale and resolution of Roskilde

Outcomes DNB and the FSC converted DKK 8.7 billion of debt issued under the facility into capital in 2009 and 2010;

(continued)

³ Per OFX, USD 1 = DKK 4.72 on March 31, 2008.

onal capital after Moody's ratings downgrade but concluded that such a capital increase could not be underwritten⁴ (Dow Jones 2008a; EC 2008a). Roskilde's board said it feared a bank run and thus sought assistance from the Danish central bank, Danmarks Nationalbank (DNB) (AFP 2008; DNB 2009b; EC 2008a).

Outcomes (continued)	the Danish state's loss from the Roskilde rescue was DKK 8.9 billion at end-2010, before interest and asset sales
Notable Features	The facility was secured by state and private guarantees, rather than collateral; the liquidity facility was conditioned on the full or partial sale of Roskilde

From July 7 to 10, 2008, Roskilde's management met with the Danish Financial Supervisory Association (DFSA) and negotiated with potential acquirers to no avail (DFSA 2009; DNB 2008a; Roskilde Bank 2008). On July 10, 2008, DNB announced it would "provide the necessary liquidity" to Roskilde (DFSA 2009, 26; DNB 2008a). DNB later confirmed that this was essentially an unlimited credit facility (EC 2008c; DNB 2023; Dow Jones 2008a). Also on July 10, DNB said that the government and private sector had agreed to cover any losses DNB might bear from the facility (DNB 2008a). The Danish Private Contingency Association (PCA) later agreed to guarantee the first DKK 750 million of potential losses on the liquidity facility, with the government responsible for any losses beyond that initial amount (EC 2008a; NAO 2009; DFSA 2009). The agreement between DNB and Roskilde contained multiple conditions, including the full or partial sale of Roskilde (AFP 2008; DNB 2009b). On July 10, 2008, Roskilde put itself up for sale (Dow Jones 2008a; Reuters 2008).

Figure 1: Annual Growth of Total Credit in Large Commercial Banks in Denmark (%)

	2003	2004	2005	2006	2007
Roskilde	13	39	55	59	37
Danske Bank	n/a	11	20	15	25
Nordea	0	10	17	-3	14
Jyske Bank	-33	17	22	18	25
Sydbank	14	22	28	20	3
Amagerbanken	-4	16	59	38	36

Source: Sjögren and Iversen 2019.

On August 24, 2008, DNB announced that with PCA it had acquired Roskilde to avoid liquidation, as Roskilde had not received any private offers (DNB 2008b; DNB 2009b; McLaughlin and Levring 2008). The PCA converted its guarantee into an equity stake of DKK 750 million (EC 2008a; EC 2011). To effectively accomplish the takeover of Roskilde, DNB and PCA established an entity called "New Roskilde" and transferred to it all assets, depositors, and other senior liabilities of "Old Roskilde," including the DKK 14.5 billion outstanding on the credit facility (EC 2008b; FSC 2010a). To ensure that New Roskilde was adequately capitalized, DNB and PCA made a capital injection of DKK 4.5 billion into New

⁴ By July 11, 2008, Roskilde's shares had fallen by 78% since the beginning of the year—six times as much as the OMX Copenhagen 20 Index, which was down by 13% (EC 2008a). The OMX Copenhagen 20 is the leading share index for the Copenhagen Stock Exchange (Nasdaq n.d.).

Roskilde. Subordinated liabilities and equity remained behind in Old Roskilde, which began insolvency proceedings (EC 2008b).

In September 2008, three banks agreed to buy 21 of Roskilde's branches (Dow Jones 2008b). The three banks took over DKK 9.3 billion in loans and DKK 4.9 billion in deposits in total. However, the three banks returned loans and deposits that they did not want to take over, at book value (EC 2011; Roskilde Bank 2009). This prolonged the Financial Stability Company's (FSC's) resolution of Roskilde longer than expected; the FSC was a newly established body owned by the Ministry of Economic and Business Affairs (EC 2011; FSC 2010a).

In October 2008, the Danish Parliament created the FSC as a more permanent means of resolving banks (Danish Parliament 2008b, pt. 2; NAO 2009; OECD 2009). In August 2009, the FSC took over New Roskilde and the credit facility from DNB (Bjerre-Nielsen 2011; IMF 2014). DNB and the FSC ultimately converted DKK 8.7 billion of debt from the credit facility into capital; including that capital, the Danish state's total investment in Roskilde Bank's equity and subordinated debt reached DKK 12.4 billion at the end of 2010 (DNB 2009b; FSC 2011b).

At the end of 2010, the Danish state's outstanding loss on the restructuring of Roskilde Bank stood at DKK 8.9 billion, not including interest and commissions earned and any later gains on asset sales (FSC 2011b). By September 2014, the FSC no longer had customer liabilities from Roskilde (FSC 2015). It is unclear if asset sales allowed the FSC to recoup some of those losses over time, or whether the state's remaining loans to the bank were fully repaid. The FSC reported positive earnings throughout the 2010s based on interest income and contributions from the PCA and member banks (MIBFA n.d.).

In December 2013, the FSC settled with the Consumer Ombudsman on the sale of bank shares by Old Roskilde to 17,000 private customers; the FSC offered these customers compensation equivalent to 60% of their net loss (see Key Design Decision No. 2, Part of a Package) (Danish Consumer Ombudsman 2013; FSC 2014). Appointed by the minister for growth and business, the Consumer Ombudsman was responsible for enforcing the Danish Marketing Practices Act which provided for fair marketing practices and consumer interests (Danish Parliament 2017, pts. 7, 9).

See Figure 2 for a timeline of major events in Roskilde's emergency liquidity assistance and restructuring.

Figure 2: Timeline of Roskilde's Emergency Liquidity Assistance and Restructuring

Date	Event
July 1, 2008	Moody's downgrades ratings of Roskilde because of the bank's exposure to real estate in a declining market and weakening asset quality.
July 10, 2008	Roskilde announces extraordinary losses and puts itself up for sale per agreement with the government. Roskilde's shares fall 57% after the announcement. DNB announces it will provide Roskilde with the necessary liquidity. The Treasury guarantees any losses that DNB might suffer related to the emergency liquidity facility provided to Roskilde.
July 15, 2008	Moody's further downgrades Roskilde's Bank Financial Strength Rating, subordinated debt, and preferred stock because of Roskilde's real estate exposure and weakening asset quality.
Aug. 24, 2008	DNB and PCA acquire Roskilde, transferring its assets and senior liabilities to a newly created entity called "New Roskilde," leaving subordinated liabilities and equity behind in "Old Roskilde," which enters bankruptcy proceedings.
Sept. 4, 2008	The Finance Committee in parliament passes Appropriation 199 4/9 2008 concerning the state guarantee for any loss suffered by DNB for the unlimited credit facility.
Sept. 25, 2008	Danish authorities hold meetings with four potential buyers of New Roskilde's assets. One buyer withdraws. The other three buyers begin negotiating specific contractual terms.
Sept. 29, 2008	Nordea, Spar Nord, and Arbejdernes Landsbank agree to buy 21 of New Roskilde's branches (see Key Design Decision No. 8, Approach to Resolution and Restructuring).
Oct. 6, 2008	Danish authorities approve the transfer of assets and senior liabilities from Old Roskilde to New Roskilde, retroactive as of August 24, 2008. DNB and PCA make a capital injection of DKK 4.5 billion into New Roskilde.
Jan. 30, 2009	DNB converts DKK 4 billion of its loans to Roskilde Bank and DKK 1 billion of its subordinated loans to the bank into equity to recapitalize Roskilde Bank.
Aug. 10, 2009	The FSC takes control of New Roskilde from DNB for winding up; PCA's stake is written off.
Dec. 30, 2009	FSC converts DKK 2 billion of its outstanding loans to the bank into equity to recapitalize Roskilde Bank; the Danish state's loss on the rescue stands at DKK 6.6 billion.
June 30, 2010	FSC converts DKK 2.7 billion of its outstanding loans to the bank into equity to recapitalize Roskilde Bank.
Dec. 31, 2010	The Danish state's loss on the rescue stands at DKK 8.9 billion.
Jan. 27, 2011	By January 27, the FSC submits an updated liquidation plan for New Roskilde to the European Commission.
June 2011	FSC restructures Roskilde assets and liabilities into FS Finans, a subsidiary of the FSC.
Dec. 2013	FSC settles with the Consumer Ombudsman on the sale of Old Roskilde's shares to private customers and offers compensation equivalent to 60% of net losses.

Source: Author's analysis.

Summary Evaluation

According to DNB, Roskilde "experienced greater withdrawal of deposits than usual" after the announcement of central bank liquidity support on July 10, 2008 (DNB 2009a, 59).

Context: Denmark 2007–2008	
GDP (SAAR, nominal GDP in USD)	\$320.2 billion in 2007
	\$355.0 billion in 2008
GDP per capita (SAAR, nominal GDP in USD)	\$58,631.7 in 2007
	\$64,616.7 in 2008
Sovereign credit rating (long-term debt)	Moody's: Aaa in 2007
	S&P: AA+ in 2007
	Fitch: AAA in 2007
	Moody's: Aaa in 2008
	S&P: AA+ in 2008
	Fitch: AAA in 2008
Size of banking system	\$658.4 billion in 2007
	\$777.7 billion in 2008
Size of banking system as a % of GDP	205.6% in 2007
	219.1% in 2008
Size of banking system as a % of financial system	Data not available in 2007
	Data not available in 2008
Five-bank concentration of banking system	87.7% in 2007
	89% in 2008
Foreign involvement in banking system	18% in 2007
	18% in 2008
Existence of deposit insurance	Yes in 2007
	Yes in 2008

Sources: Bloomberg; World Bank Deposit Insurance Dataset; World Bank Global Financial Development Database.

Key Design Decisions

- 1. Purpose: DNB and PCA agreed to provide a credit facility until Roskilde could be fully or partially sold. When Danish authorities created a bridge bank, they used the credit facility to redeem all senior creditors.**

Roskilde had considerable exposure to real estate and construction firms in July 2008 (Moody's 2008). Rating agencies downgraded Roskilde when the bank's management discovered that Roskilde would have to take larger write-downs than expected (AFP 2008; DNB 2009b; EC 2008a; NAO 2009; Dow Jones 2008a). Later reports reveal that Roskilde's management initially told Danish authorities that the bank was solvent, but they feared a bank run and thus sought assistance from DNB (Bernstein 2008; DNB 2009b; Dow Jones 2008a; EC 2008a). DNB and PCA agreed to grant an unlimited credit facility, conditioned on Roskilde's full or partial sale to simultaneously avoid speculation about suspension of payments, and avoid setting the precedent of a governmental safety net for other distressed banks (AFP 2008; DNB 2009b; EC 2008a; NAO 2009). On July 10, 2008, Roskilde put itself up for sale (Dow Jones 2008a; Reuters 2008).

On August 22, 2008, external auditors told the DFSA that Roskilde did not comply with the individually calculated capital need or the statutory solvency requirement. The DFSA granted Roskilde regulatory relief until August 29, 2008, to meet its individual capital need and statutory solvency requirement; the DFSA subsequently extended this relief to September 16, 2008 (DNB 2008c).

Then-DNB Governor Nils Bernstein spoke about Roskilde at a press conference on August 27, 2008:

We find that the circumstances of Roskilde Bank now pose a considerable threat to financial stability in Denmark. An important factor in this context is that several other Danish banks are also relying on the international capital markets as a source of financing. Consequently, any losses on loans to Roskilde Bank may have an adverse knock-on effect on other Danish banks. (Bernstein 2008)

Roskilde could only use the credit facility to cover liquidity that was necessary for the bank's day-to-day activities. Roskilde could not use the facility to invest in new activities, take on risky positions in commercial markets, engage in new, more extensive credit facilities, or actively market and attract new depositors or borrowers (EC 2008a).

When DNB created a new bridge bank (New Roskilde) on August 24, 2008, DNB said that it used the credit facility to repay all senior liabilities of New Roskilde. Danish authorities said there was a legal obligation to pay these creditors for two reasons. First, the creditors' contracts had reached maturity, and the creditors did not want to extend their contracts. Second, the transfer of assets and liabilities from Old Roskilde to New Roskilde constituted an event of default, and the remaining creditors thus had the right to terminate their

contracts and demand repayment. All creditors chose redemption rather than transfer to New Roskilde⁵ (EC 2008b).

2. Part of a Package: The government announced a broad-based two-year guarantee for all bank deposits. DNB and PCA injected capital into the bridge bank, New Roskilde.

Broad-Based Deposit Guarantee

Effective October 10, 2008, Minister for Economic Affairs Lene Espersen announced a two-year General Guarantee Scheme for all bank deposits,⁶ excluding deposits at Roskilde (FSC 2011a; FSC 2011b; Levring 2008). However, an agreement between the FSC and the PCA signed on August 10, 2009, provided DKK 10.5 billion from the General Guarantee Scheme to cover accounts at Roskilde (FSC 2010a; FSC 2011a).

Capital Injections

Following the acquisition of Roskilde on October 6, 2008, DNB and PCA made a capital injection of DKK 4.5 billion into Roskilde to meet solvency requirements (EC 2008b; IMF 2014). DNB injected DKK 1.7 billion in the form of class A shares and DKK 2.0 billion in the form of subordinated debt that counted as capital; PCA injected DKK 750 million in the form of class B shares. The class B shares ensured that Roskilde's incurred losses would first be covered by the PCA before DNB (EC 2008b; EC 2011).

Resolution

On August 24, 2008, DNB announced that it and the PCA acquired Roskilde to avoid liquidation, as Roskilde had not received any private offers (see (Decker 2024)) (DNB 2008b; DNB 2009b; McLaughlin and Levring 2008).

In December 2013, the FSC settled with the Consumer Ombudsman on the sale of bank shares by Old Roskilde. The settlement concerned private customers who bought shares in connection with two share sales campaigns: one from August 11, 2006, to September 17, 2006, and another from March 12, 2007, to April 8, 2007. The FSC identified 17,000 private customers that were entitled to damages. The FSC offered these customers compensation equivalent to 60% of their net loss (Danish Consumer Ombudsman 2013; FSC 2014). According to the Consumer Ombudsman, the compensation took into account the general risks involved in buying Roskilde's shares and the additional risk compared to other banks' shares. If shareholders did not want to accept this offer, the settlement said that the limitation period was suspended for 24 weeks after the settlement's conclusion. Thus, shareholders had the opportunity to submit a complaint to the FSC or the Bank Appeals

⁵ According to Roskilde's annual report, the bank had DKK 2.1 billion in deposits as of December 2008 (Roskilde Bank 2009).

⁶ The General Guarantee Scheme supplemented an existing, privately funded deposit insurance system administered by DNB, which covered deposits of all PCA members up to EUR 50,000 (DGFDI 2009; PCA 2011). For more on Denmark's General Guarantee Scheme, see Hoffner (2022).

Board, or shareholders could file a court case. The settlement exempted Roskilde's senior employees, customer advisors, and others with special knowledge of the bank's conditions or special knowledge of the risk of buying shares (Danish Consumer Ombudsman 2013).

3. Legal Authority: DNB's bylaws allowed its board of directors to authorize matters of special importance, and the Treasury received approval from parliament to guarantee losses on the credit facility.

Danish Legal Authority

DNB could grant advances against security, according to Section 22 of the central bank bylaws (Danish Parliament 1936). Additionally, according to Section 4 of DNB's bylaws, the DNB board of directors decided "all matters of special importance which may be considered to be outside the domain of daily business" (Danish Parliament 1936, 117). While no primary source documentation refers to the DNB articles of association as the legal basis of the emergency liquidity in question, the author notes Sections 4 and 22 as likely sources of legal authority.

The Treasury needed approval from the Finance Committee of the Danish Parliament to guarantee losses beyond the PCA's commitment. The minister for economic and business affairs proposed legislation on July 13, 2008, which was revised on July 18, 2008 (EC 2008a). On September 4, 2008, the Finance Committee passed Appropriation 199 4/9 2008 concerning the state guarantee for any loss suffered by DNB for the unlimited credit facility. The Finance Committee passed an additional appropriation, Appropriation 25 27/11 2008, on November 4, 2008, concerning DNB's part in the unlimited credit facility (NAO 2009).

As a member of the EU, Denmark must comply with the prohibition on monetary financing contained in Article 123(1) of the Treaty on the Functioning of the European Union (TFEU)⁷, which reads (emphasis added):

Overdraft facilities or **any other type of credit facility** with the European Central Bank or with the central banks of the Member States (hereinafter referred to as "national central banks") in favour of Union institutions, bodies, offices or agencies, central governments, regional, local or other public authorities, other bodies governed by public law, or **public undertakings of Member States** shall be prohibited, as shall the purchase directly from them by the European Central Bank or national central banks of debt instruments. (EU 2009, art. 123)

⁷ The TFEU, itself based on the Treaty of Maastricht of 1992, is one of the EU's two foundational treaties (the other being the Treaty on European Union). It sets out in detail the principles and objectives of the EU and the scope and functional responsibilities of EU institutions (the Treaty on European Union in contrast defines the EU's purpose and lays out its institutions and form). As a technicality, before the Lisbon Treaty came into effect in 2009, TFEU Article 123 was Article 101 of the Treaty Establishing the European Community, but the content was unchanged when the Lisbon Treaty relocated the article.

TFEU Article 123 prohibits the extension of liquidity to insolvent firms (at least without a state guarantee), which is considered a public undertaking (in other words, one that should be supported with state resources). The monetary financing prohibition—including its application to emergency liquidity—is EU-level law, and thus applies even to those nations that are Member States of the EU but not members of the Eurosystem. For more on the application of the monetary financing prohibition to ELA, see (Arnold 2025).

EC Decision-Making

On July 31, 2008, the European Commission (EC) ruled that the part of the unlimited credit facility guaranteed by the PCA did not constitute State Aid, while the part of the unlimited credit facility guaranteed by DNB did constitute State Aid under the application of Article 87(3)(c) of the EC Treaty and the Community guidelines on State Aid for rescuing and restructuring firms in difficulty. On compatibility, the EC said that Roskilde met the five conditions of the community guidelines. First, the assistance mostly took the form of loans and loan guarantees. Furthermore, the interest rate was higher than Denmark's reference rate and the duration of the facility could be terminated at any time by DNB. Second, the aid was for the purpose of preventing serious negative social and economic consequences of a run on Roskilde which could have led to contagion. The EC also said that this aid would not have undue adverse effects on other member states. Third, Danish authorities committed to providing the following within six months: (a) restructuring plan, or (b) liquidation plan, or (c) proof that the facility was repaid in full, or (d) proof that the guarantee was terminated. Fourth, the facility would allow Roskilde only the minimum amount necessary of liquidity, verified by an independent auditor; Roskilde was not in a position to act aggressively in the market. Fifth, this was the first instance of Roskilde receiving rescue or restructuring aid (EC 2008a).

On August 24, 2008, Danish authorities said that they were not aware of the European Central Bank's strict interpretation of monetary financing rules for insolvent banks, since DNB's risk was covered by the state's unlimited guarantee (NAO 2009).

In its subsequent decision in November 2008 on Roskilde's resolution, the EC said that it had observed a "substantial change in circumstances" and needed to reevaluate the ad hoc liquidity facility (EC 2008b, 13). The EC noted that the unlimited credit facility originally granted to Roskilde on July 10, 2008, was transferred to New Roskilde on October 6, 2008, per the resolution plan.⁸ The EC concluded that the unlimited credit facility, as part of a package granted to Roskilde (also including recapitalization, restructuring, and resolution measures), constituted liquidation aid rather than restructuring aid. Therefore, the question was whether this liquidation aid was compatible with the common market. The EC ultimately decided that the unlimited credit facility constituted State Aid compatible with Article 87(3)(b) of the EC Treaty (EC 2008b). The package of aid to Roskilde was "necessary to remedy a serious disturbance of [the] Danish economy in the current market situation [by

⁸ Since the transfer, New Roskilde had made further drawdowns on the facility, and Danish authorities indicated to the EC that even more drawdowns were forthcoming in order to pay senior creditors who had demanded immediate reimbursement (see Key Design Decision No. 7, Source and Size of Funding) (EC 2008b).

allowing] the continuation of the senior debt funding among financial institutions” (EC 2008b, 19). In its justification, the EC noted that “a private sector solution was first tried without success before committing any additional state resources” (EC 2008b, 20).

4. Administration: The unlimited credit facility was coordinated among the PCA, Treasury, and DNB.

Roskilde’s board said it feared a bank run and thus sought assistance from DNB on July 10, 2008 (AFP 2008; DNB 2009b; EC 2008a; Roskilde Bank 2008). DNB consulted the ministry of economic and business affairs, DFSA, and DBA before granting emergency liquidity assistance, while Roskilde was readying itself for sale (AFP 2008; DNB 2009b; ENP 2008; EC 2008c; NAO 2009). Danish officials later said that they chose DNB as the administrator of the unlimited credit facility, because it was the central bank’s job to support temporarily illiquid but solvent banks (NAO 2009).

DNB administered the liquidity facility (EC 2008a). DNB was also responsible for Roskilde’s administration and related costs after the government took over the distressed institution on August 24, 2008; the PCA itself could not take over a distressed bank, according to its own articles of association (NAO 2009; PCA 2011). DNB received weekly summaries of its account with Roskilde. DNB regularly notified the minister for economic and business affairs and the Finance Committee of the Danish Parliament about losses absorbed by the government (NAO 2009).

The PCA’s initial guarantee of DKK 750 million was an indemnity bond. The PCA would make its payment under the guarantee as soon as Roskilde failed to repay the loan to DNB, regardless of whether Roskilde underwent insolvency proceedings (EC 2008a). Though it was possible to double the PCA’s contribution to DKK 1.5 billion according to the consortium’s bylaws, the government did not pursue this option because it could not obtain approval from all PCA members in the necessary timeframe (NAO 2009).

The Danish government and DNB chose PricewaterhouseCoopers (PwC) to verify and approve the cashflow budget required for drawdowns of the credit facility (EC 2008a).

The PCA was funded by voluntary contributions from member banks and from the Danish firm Nykredit (DNB 2008b; NAO 2009; PCA 2011). Member banks contributed a maximum annual contribution of up to DKK 750 million per year, unless otherwise unanimously decided by the board of representatives⁹ (EC 2008a; PCA 2011). The PCA guaranteed a maximum of DKK 1.5 billion per calendar year; the board of representatives could change this amount with a unanimous vote (EC 2008a; NAO 2009; PCA 2011).

The FSC took over New Roskilde and its credit facility from DNB in August 2009 (FSC 2010a; FSC 2011b). In its annual report for 2010, the FSC indicated that New Roskilde could still

⁹ For perspective, the size of Denmark’s gross domestic product was USD 353.4 billion in 2008 (per the World Bank).

draw on the transferred facility at that time (see Key Design Decision No. 7, Source and Size of Funding) (FSC 2011b).

The FSC reported outstanding loans of DKK 4.4 billion to New Roskilde Bank at the end of 2010, including the legacy DKK 1 billion subordinated debt and DKK 3.4 billion from “ordinary debt financing,” presumably representing debt under the loan facility (FSC 2011b, 29; FSC 2012). Even after New Roskilde’s assets and liabilities were subsumed by FS Finans and FS Bank in June 2011, the FSC reported outstanding loans of DKK 1.9 billion to “FS Finans A/S, Roskilde,” presumably still including the subordinated loan and the legacy credit facility (FSC 2011b; FSC 2012).

5. Governance: PwC audited usage of the credit facility, and DNB’s ad hoc decisions were decided by its board of directors.

PwC verified and approved the cashflow budget for drawdowns of the unlimited credit facility. PwC particularly scrutinized whether the budgeted needs complied with the behavioral conditions attached to the credit facility (for example, no investments in new activities) (EC 2008a).

DNB’s board of directors decided on important questions outside of normal, daily central bank operations. As a decision-making body, the DNB board of directors comprised 25 members: eight members elected from parliament, one of whom resigns their seat in parliament; one economist; one lawyer; and 15 members “with a thorough knowledge of the economy” who were elected by other members of the DNB board of directors (Danish Parliament 1936, 116). The board internally elected a chair and deputy chair for one-year terms. The board made decisions by majority vote, and the chair was the deciding vote in the event of a tie (Danish Parliament 1936).

6. Communication and Disclosure: Danish authorities said that they would provide the necessary liquidity to Roskilde.

On July 10, 2008, DNB issued a press release stating that it would “provide the necessary liquidity to Roskilde Bank A/S. The Government and the financial sector have assured [DNB] of their support to the arrangement with a guarantee towards [DNB]” (DNB 2008a). The press release also said that the “amount and scope of [Roskilde’s] write-downs are not known at this point but will be made public once the full amount is known” (DNB 2008a).

On June 17, 2009, the DFSA released its own report on the causes of Roskilde’s collapse. The DFSA also self-assessed its response. Usually, such reports could not be made public by the DFSA or minister for economy and business, according to Section 354 of the Financial Business Act (DFSA 2009; Danish Parliament 2008a, sec. 354). However, according to the Financial Business Act, Section 352(a)1–2, the DFSA must prepare a public report in the following circumstances: (a) a financial institution receives state funding or guarantees, or (b) a financial institution has been declared bankrupt. Thus, the DFSA was allowed to publicize its findings because DNB had provided a liquidity facility to Roskilde, guaranteed by the government, and Roskilde had been declared bankrupt (DFSA 2009).

7. Source and Size of Funding: The credit facility was unlimited and was funded by DNB's creation of central bank reserves.

The Danish government said it was DNB's responsibility to cultivate support for the guarantee with the parliament's Finance Committee. On September 4, 2008, the Finance Committee approved Appropriation 199 4/9 2008 for the state's guarantee on any losses incurred by DNB's unlimited credit facility to Roskilde. The Finance Committee later approved Appropriation 25 27/112008, which expanded the government's guarantee to include any losses incurred in the takeover and resolution of Roskilde (NAO 2009). (See Decker [2024] for more information on Roskilde's restructuring and resolution.)

On July 18, 2008, the minister for economic and business affairs told the Finance Committee of the Danish Parliament that he did not expect drawdowns on the unlimited credit facility to exceed DKK 15–20 billion. However, after Lehman Brothers failed and led to market stress across the world in addition to the announcement of the government takeover, the EC reported that Roskilde drew down DKK 36.8 billion on the facility by early October on a short-term basis to allow the bank to pay depositors and other senior creditors by the end of that month (see Key Design Decision No. 1, Purpose). Danish authorities said that Roskilde's creditors were legally allowed to demand immediate repayment for two reasons: (a) their contracts had matured, and these creditors did not want to extend their contracts, and (b) the impending transfer of Roskilde's assets and liabilities to a new legal entity constituted an event of default (EC 2008b). New Roskilde continued to draw on emergency liquidity assistance until the FSC merged New Roskilde's assets into a liquidation vehicle in 2011 (FSC 2011b; FSC 2012).

Research did not uncover the Danish state's total losses on the credit facility. According to an EC document in November 2008:

The possible losses of the DNB (as a creditor and equity owner in [Roskilde]) will only be quantifiable once the new bank has been finally wound-up. They will essentially depend on the value of [Roskilde's] loans and bonds [...] or of the underlying assets serving as collateral (EC 2008b, 7).

At the end of 2010, the Danish state's outstanding loss on the restructuring of Roskilde stood at DKK 8.9 billion, although it is possible the FSC may have recovered a portion of that through interest and commissions earned and any later gains on asset sales. Since DKK 8.7 billion of the state's DKK 12.4 billion total investment in New Roskilde came from the conversion of emergency liquidity support to equity, the majority of that loss for the Danish state came from the credit facility (DNB 2009b; FSC 2010b; FSC 2011b).

It is unclear whether the state's remaining loans to the bank were fully repaid or whether FSC has remaining legacy assets from Roskilde. The FSC reported positive earnings throughout the 2010s based on interest and commission income from member banks (MIBFA n.d.).

Figure 3 shows the usage of the liquidity facility through December 2010.

Figure 3: Liquidity Facility Usage and Conversion to Equity, DKK millions

	31 Dec. 2008	30 Jan. 2009	10 Aug. 2009	31 Dec. 2009	31 Dec. 2010
Liquidity facility used	17,517	13,517	14,355	10,578	3,383
Total DNB/FSC equity exposure	1,723	6,723	6,723	8,723	11,423
DNB/FSC sub. debt exposure	2,000	1,000	1,000	1,000	1,000
Total DNB/FSC exposure	21,240	21,240	22,078	20,301	15,806
Liquidity facility available	36,800	32,800	32,800	30,800	28,100
Liquidity facility unused	19,283	19,283	18,445	20,222	24,717

Sources: DNB 2009b; DNB 2010; EC 2011; FSC 2010a; FSC 2011b; author's analysis.

8. Rates and Fees: The interest rate was 6.6%, and Roskilde did not have to pay any fees for the credit facility.

The credit facility's rate was 200 basis points (bps) above DNB's ordinary rate for loans and 100 bps above the rate for DNB's standing facility for seven-day liquidity assistance. As of July 31, 2008, DNB's ordinary rate for loans was 4.6% and the standing facility's rate was 5.6%, meaning that the rate for Roskilde's unlimited credit facility was 6.6% (EC 2008a).

Neither the PCA nor the state charged fees for their guarantees of the credit facility (EC 2008a).

9. Loan Duration: The credit facility did not have a termination date, but DNB initially committed to finding a solution for Roskilde within six months.

DNB could terminate the credit facility at any time without notice. While the loan agreement did not have a termination date, DNB committed to finding a solution for Roskilde within six months. By January 11, 2009, DNB had to provide a restructuring plan, liquidation plan, or proof that the credit facility was completely repaid with termination of the state guarantee (EC 2008a).

On August 24, 2008, since the bank did not receive any offers after putting itself up for sale, DNB and the PCA acquired Roskilde to avoid liquidation (Decker 2024; DNB 2009b; McLaughlin and Levring 2008). Danish authorities chose to create New Roskilde because taking assets and liabilities from Old Roskilde directly would have entailed expropriation and a complicated legal process that Danish authorities did not deem appropriate (EC 2008b). The PCA would convert its guarantee into an equity stake of DKK 750 million (DNB 2008b; EC 2008a; EC 2011). According to DNB's third quarter report for 2008, restructuring and capital injections from DNB and PCA would best allow Roskilde to settle its activities (DNB 2008c).

The FSC reported outstanding loans of DKK 4.4 billion to New Roskilde at the end of 2010, including the legacy DKK 1 billion subordinated debt and DKK 3.4 billion from "ordinary debt financing," presumably representing debt under the loan facility (FSC 2011b; FSC 2012). Even after New Roskilde's assets and liabilities were subsumed by FS Finans and FS Bank in

June 2011, the FSC reported outstanding loans of DKK 1.9 billion to “FS Finans A/S, Roskilde,” presumably still including the subordinated loan and the legacy credit facility (FSC 2011b; FSC 2012).

10. Balance Sheet Protection: Roskilde did not provide collateral, and the facility was guaranteed by the PCA and the state.

Roskilde was unable to provide sufficient collateral, with the liquidity facility secured by the PCA’s guarantee of DKK 750 million and the state’s guarantee beyond that initial amount; this meant that DNB was to be protected from losses on the facility (see Key Design Decision No. 1, Purpose) (DNB 2009b; EC 2008a). Initially, Danish authorities anticipated drawdowns on the facility between DKK 15 billion–DKK 20 billion. The state’s guarantee would only be triggered after Roskilde’s equity and subordinated capital and the PCA’s guarantee were exhausted (EC 2008a).

Private sector support from the PCA reflected negotiations among DNB, PCA, and the ministry of economic and business affairs. While no calculations were performed for private sector support, the PCA’s contribution could not exceed the government’s estimated loss if Roskilde were to enter bankruptcy, following the PCA’s articles of association (NAO 2009; PCA 2011). The PCA wrote off its DKK 750 million investment in Roskilde after the government’s acquisition (FSC 2010a).

It became apparent that Roskilde had to make further write-downs that would prevent the bank from meeting the calculated capital adequacy need and capital requirement of the Financial Services Act.¹⁰ On August 29, 2008, the chairman of the Finance Committee received a letter from the ministry of economic and business affairs about additional appropriations needed. The ministry thought that Appropriation 199 could not be adjusted, because the EC had already approved the measures necessary for the emergency liquidity assistance, and these measures had already been taken. Thus, the Finance Committee had to pass additional appropriations. On November 4, 2011, the Finance Committee passed Appropriation 25 27/11 2008 to extend the government’s guarantee to include any losses incurred in connection with DNB’s takeover and resolution of Roskilde (NAO 2009).

11. Impact on Monetary Policy Transmission: Research has not uncovered any impact on monetary policy transmission.

Research did not uncover any impacts from the unlimited credit facility on monetary policy transmission in Denmark.

¹⁰ An institution’s base capital had to amount to a minimum of 8% of the institution’s risk-weighted assets or capital adequacy need (NAO 2009).

12. Other Conditions: Roskilde had to commit to a sale to receive emergency liquidity assistance.

The unlimited credit facility was conditioned on Roskilde's full or partial sale (DNB 2008a; DNB 2009b).

References and Key Program Documents

Implementation Documents

(Bjerre-Nielsen 2011) Bjerre-Nielsen, Henrik. 2011. "Letter to Anders Aagaard." June 12, 2011.

Letter from the Danish Financial Stability Company CEO Henrik Bjerre-Nielsen to Anders Aagaard at Norsker & Co Advokater about Roskilde's financial results.

<https://elischolar.library.yale.edu/ypfs-documents2/983>

(PCA 2011) Private Contingency Association (PCA). 2011. Articles of Association. May 9, 2011.

Articles of Association for the Private Contingency Association, discussing the administration and terms of the PCA (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents/14724>

Legal/Regulatory Guidance

(Danish Parliament 1936) Danish Parliament. 1936. Act on Danmarks Nationalbank. Act no. 116 of July 4, 1936.

Law setting up Denmark's central bank to maintain a secure monetary system (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents2/1005>

(Danish Parliament 2008a) Danish Parliament. 2008a. Promulgation of the Financial Business Act. LBK No. 376 of May 22, 2008.

Law allowing the transfer of Roskilde's activities to the government (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents2/1009>

(Danish Parliament 2008b) Danish Parliament. 2008b. Law on Financial Stability. Law No. 1003 of October 10, 2008.

Law passed on October 10, 2008, which introduced the General Guarantee Scheme (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents/4070>

(Danish Parliament 2017) Danish Parliament. 2017, Danish Marketing Practices Act. *Law providing for the Consumer Ombudsman in Denmark (translated to English).*

<https://elischolar.library.yale.edu/ypfs-documents2/4407>

(EC 2008a) European Commission (EC). 2008a. "State Aid NN 36/2008 – Denmark - Roskilde Bank A/S." July 31, 2008.

Decision by the EC as to whether liquidity support to Roskilde constituted State Aid.

<https://elischolar.library.yale.edu/ypfs-documents2/1006>

(EC 2008b) European Commission (EC). 2008b. "State Aid NN 39/2008 – Denmark – Aid for Liquidation of Roskilde Bank." November 5, 2008.

Decision by the EC to not object to Denmark's State Aid measures.

<https://elischolar.library.yale.edu/ypfs-documents2/1007>

(EC 2011) European Commission (EC). 2011. "NN 52/2010 – Denmark – Amendments to Liquidation Aid to Roskilde Bank." May 24, 2011.

Decision by the EC approving amendments to Roskilde's liquidation plan.

<https://elischolar.library.yale.edu/ypfs-documents2/1008>

(EU 2009) European Union (EU). 2009. Treaty on the Functioning of the European Union. 5/8/2008.

Treaty prohibiting monetary financing by Eurosystem national central banks.

<https://elischolar.library.yale.edu/ypfs-documents2/3841>

Media Stories

(AFP 2008) Agence France Presse (AFP). 2008. "Danish Central Bank Rescues Roskilde Bank from Subprime Woes." Agence France Presse, July 11, 2008.

News article on DNB's support to Roskilde, conditioned on the full or partial sale of the bank.

<https://elischolar.library.yale.edu/ypfs-documents2/1010>

(Dow Jones 2008a) Dow Jones. 2008a. "Roskilde Bank Puts Itself up for Sale as Write-Downs Hit." Dow Jones, July 11, 2008.

News article on DNB's assistance to Roskilde, conditioned on its full or partial sale.

<https://elischolar.library.yale.edu/ypfs-documents2/1014>

(Dow Jones 2008b) Dow Jones. 2008b. "Roskilde Bank: Nordea, Spar Nord Bank, Arbejdernes Landsbank to Buy Roskilde Branches." September 29, 2008.

News article containing high-level information on the acquisition of Roskilde branches by three banks.

<https://elischolar.library.yale.edu/ypfs-documents2/1016>

(ENP 2008) Electronic News Publishing (ENP). 2008. "Danmarks Nationalbank – The Danish Financial Supervisory Authority – Report." Electronic News Publishing, July 11, 2008.

News article on Roskilde's larger-than-expected write-downs.

<https://elischolar.library.yale.edu/ypfs-documents2/1017>

(FT 2008) Financial Times (FT). 2008. "Bank Failures: Roskilde." August 26, 2008.

Lex opinion column on Roskilde's failure.

<https://elischolar.library.yale.edu/ypfs-documents2/1034>

(Levring 2008) Levring, Peter. 2008. "Update 1–Interview–Danish Economy Minister Says Banks Are Strong." Reuters, October 6, 2008.

News article containing quotes from Minister Espersen in October 2008.

<https://elischolar.library.yale.edu/ypfs-documents2/1037>

(McLaughlin and Levring 2008) McLaughlin, Kim, and Peter Levring. 2008. "Update 3–Danish Central Bank Buys Out Ailing Roskilde Bank." Reuters, August 25, 2008.
News article containing characterizations by the DNB governor of Roskilde's bailout and takeover.

<https://elischolar.library.yale.edu/ypfs-documents2/1038>

(Reuters 2008) Reuters. 2008. "Update 2–Danish Bank Roskilde Hits Trouble, for Sale." Reuters, July 11, 2008.

News article on Roskilde's bailout, conditioned on its full or partial sale.

<https://elischolar.library.yale.edu/ypfs-documents2/1040>

Press Releases/Announcements

(Bernstein 2008) Bernstein, Nils. 2008. "Actions Taken over Roskilde Bank." Introductory statement delivered at a press conference at Danmarks Nationalbank, Copenhagen, Denmark, August 27, 2008.

Statement by the DNB governor at a press conference about Roskilde.

<https://elischolar.library.yale.edu/ypfs-documents2/1042>

(DNB 2008a) Danmarks Nationalbank (DNB). 2008a. "The Danish Financial Supervisory Authority." Press release, July 10, 2008.

Press release from DNB announcing liquidity support to Roskilde following the bank's unexpectedly large write-downs.

<https://elischolar.library.yale.edu/ypfs-documents2/1894>

(DNB 2008b) Danmarks Nationalbank (DNB). 2008b. "Roskilde Bank." Press release, August 24, 2008.

Press release from DNB announcing the takeover of Roskilde.

<https://elischolar.library.yale.edu/ypfs-documents2/1895>

(EC 2008c) European Commission (EC). 2008c. "State Aid: The Commission Approves Danish Rescue Package for Roskilde Bank." Press release, July 31, 2008.

Press release from the EC as to whether liquidity support constituted State Aid.

<https://elischolar.library.yale.edu/ypfs-documents2/1043>

(FSC 2011a) Finansiel Stabilitet (Financial Stability Company, FSC). 2011a. "Accounts for the Bank Package." April 2, 2011.

FSC press release announcing the account types eligible for the bank package (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents/14723>

(MIBFA n.d.) Ministry of Industry, Business and Financial Affairs (MIBFA). n.d. "Final Calculation of the Surplus of the Bank Packages."

Note showing a profit of DKK 17 billion, the final status of the bank packages (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents2/1897>

(Moody's 2008) Moody's Investor Service (Moody's). 2008. "Moody's Downgrades Roskilde Bank's BFSR to C- and GLC Deposit Rating to A2." Press release, July 1, 2008.
Press release announcing Moody's downgrades of Roskilde in light of the bank's exposure to real estate and weakening asset quality.

<https://elischolar.library.yale.edu/ypfs-documents2/1050>

(Danish Consumer Ombudsman 2013) Danish Consumer Ombudsman. 2013.

"Compensation to Bank Shareholders." December 2, 2013.

Webpage detailing the settlement with the Consumer Ombudsman on the sale of Old Roskilde shares to private investors (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents2/1900>

(Nasdaq n.d.) Nasdaq. n.d. "Overview for OMXC20." Accessed February 12, 2024.

Webpage with information on the OMX Copenhagen 20 as the leading stock index on the Copenhagen stock exchange.

<https://elischolar.library.yale.edu/ypfs-documents2/4406>

Reports/Assessments

(DFSA 2009) Danish Financial Supervisory Authority (DFSA). 2009. "Statement of the Danish Financial Supervisory Authority's Supervision of Roskilde Bank Pursuant to Section 352 a of the Financial Business Act." June 17, 2009.

Report on the causes of Roskilde's failure and the process leading up to bankruptcy (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents2/1901/>

(DGFDI 2009) Danish Guarantee Fund for Depositors and Investors (DGFDI). 2009. Annual Report 2008.

Annual report for 2008 for the Danish Guarantee Fund for Depositors and Investors (in Danish).

<https://elischolar.library.yale.edu/ypfs-documents/14725>

(DNB 2008c) Danmarks Nationalbank (DNB). 2008c. "Monetary Review 3rd Quarter." 2008.

Report from DNB for the third quarter of 2008.

<https://elischolar.library.yale.edu/ypfs-documents2/4405>

(DNB 2009a) Danmarks Nationalbank (DNB). 2009a. "Financial Stability 2008: 2nd Half." January 2009.

Report from DNB for the second half of 2008.

<https://elischolar.library.yale.edu/ypfs-documents2/1053>

(DNB 2009b) Danmarks Nationalbank (DNB). 2009b. Reports and Accounts 2008.

Central bank annual report for 2008.

<https://elischolar.library.yale.edu/ypfs-documents2/529>

(DNB 2010) Danmarks Nationalbank (DNB). 2010. Reports and Accounts 2009.
Central bank annual report for 2009.
<https://elischolar.library.yale.edu/ypfs-documents2/1902>

(DNB 2023) Danmarks Nationalbank (DNB). 2023. "Monetary History of Denmark." Vol. 7, 2005–2020, January 25, 2023.
Report by DNB on the monetary history of Denmark, including financial sector reform during and after the Global Financial Crisis.
<https://elischolar.library.yale.edu/ypfs-documents2/1892>

(FSC 2010a) Finansiel Stabilitet (Financial Stability Company, FSC). 2010a. Annual Report 2009.
Annual report by the FSC for 2009.
<https://elischolar.library.yale.edu/ypfs-documents/14729>

(FSC 2010b) Finansiel Stabilitet (Financial Stability Company, FSC). 2010b. Roskilde Bank.
Report by the FSC on Roskilde for January through June 2010 (in Danish).
<https://elischolar.library.yale.edu/ypfs-documents2/4408>

(FSC 2011b) Finansiel Stabilitet (Financial Stability Company, FSC). 2011b. Annual Report 2010.
Annual report by the FSC for 2010.
<https://elischolar.library.yale.edu/ypfs-documents/14895>

(FSC 2012) Finansiel Stabilitet (Financial Stability Company, FSC). 2012. Annual Report 2011.
Annual report by the FSC for 2011.
<https://elischolar.library.yale.edu/ypfs-documents/14896>

(FSC 2014) Finansiel Stabilitet (Financial Stability Company, FSC). 2014. Annual Report 2013.
Annual report by the FSC for 2013.
<https://elischolar.library.yale.edu/ypfs-documents2/1891>

(FSC 2015) Finansiel Stabilitet (Financial Stability Company, FSC). 2015. Annual Report 2014.
Annual report by the FSC for 2014.
<https://elischolar.library.yale.edu/ypfs-documents2/1890>

(IMF 2014) International Monetary Fund (IMF). 2014. "Denmark: Financial Sector Assessment Program." IMF Country Report No. 14/351, December 2014.
IMF FSAP dated 2008 containing historical information on Roskilde during 2008.
<https://elischolar.library.yale.edu/ypfs-documents2/1059>

(Ministry of Business and Growth 2013) Ministry of Business and Growth. 2013. "The Financial Crisis in Denmark: Causes, Consequences, and Learning." September 18, 2013. *Retrospective report on the GFC, commissioned by the government (in Danish)*.
<https://elischolar.library.yale.edu/ypfs-documents2/1060>

(NAO 2009) National Audit Office (NAO). 2009. "Report to the Public Accounts Committee on the Danish Financial Supervisory Authority's Activities in Relation to Roskilde Bank A/S." June 2009.
Report by the NAO to the PAC to evaluate the role of the DFSA in Roskilde's collapse and winding-up.
<https://elischolar.library.yale.edu/ypfs-documents2/1061>

(OECD 2009) Organisation for Economic Co-Operation and Development (OECD). 2009. "Chapter 1: Denmark: The Crisis and Beyond." In *OECD Economic Surveys, 2009: Denmark*. Paris: OECD 2009, no. S1: 17–64.
OECD summary of Denmark's crisis response.
<https://elischolar.library.yale.edu/ypfs-documents/14761>

(Roskilde Bank 2008) Roskilde Bank. 2008. "Half-Year Report 2008." August 29, 2008.
Report by Roskilde for the first half of 2008.
<https://elischolar.library.yale.edu/ypfs-documents2/1886>

(Roskilde Bank 2009) Roskilde Bank. 2009. Annual Report 2008.
Annual report by Roskilde for 2008 (in Danish).
<https://elischolar.library.yale.edu/ypfs-documents2/1889>

Key Academic Papers.

(Arnold 2025) Arnold, Vincient. 2025. "Emergency Liquidity Assistance and Monetary Financing in the European Union: A Case Study in Fiscal Cooperation?" Yale Program on Financial Stability, *Systemic Risk Blog*, January 30, 2025.
Policy note discussing the application of monetary financing regulations on ELA in Europe.
<https://som.yale.edu/story/2025/emergency-liquidity-assistance-and-monetary-financing-european-union-case-study-fiscal>

(Decker 2024) Decker, Bailey. 2024. "Denmark: Roskilde Bank Restructuring, 2008." *Journal of Financial Crises* 6, no. 1.
YPFS case study examining the restructuring of Roskilde during the GFC.
<https://elischolar.library.yale.edu/journal-of-financial-crises/vol6/iss1/7/>

(Hoffner 2022) Hoffner, Benjamin. 2022. "Denmark: General Guarantee Scheme, 2008." *Journal of Financial Crises* 4, no. 4.
YPFS case study on Denmark's guarantee of deposits and other senior liabilities during the GFC.
<https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss4/6>

(Kelly et al. 2025) Kelly, Steven, Vincient Arnold, Greg Feldberg, and Andrew Metrick. 2025. "Survey of Ad Hoc Emergency Liquidity Programs." *Journal of Financial Crises* 7, no. 1. *Survey of YPFS case studies examining the provision of ad hoc emergency liquidity.*
<https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/3/>

(Sjögren and Iversen 2019) Sjögren, Hans, and Martin Jes Iversen. 2019. "The State as the Investor of Last Resort: A Comparative Study of Banking Crises in Denmark and Sweden." *Scandinavian Economic History Review* 67, no. 2: 171–89. *Article discussing bank bailouts in Sweden and Denmark.*
<https://elischolar.library.yale.edu/ypfs-documents2/148>

(Wiggins et al. 2022) Wiggins, Rosalind Z., Sean Fulmer, Greg Feldberg, and Andrew Metrick. 2022. "Broad-Based Emergency Liquidity Programs." *Journal of Financial Crises* 4, no. 2. *Survey of YPFS case studies examining broad-based emergency liquidity programs.*
<https://elischolar.library.yale.edu/journal-of-financial-crises/vol4/iss2/3/>

Copyright 2025 © Yale University. All rights reserved. To order copies of this material or to receive permission to reprint any or all of this document, please contact the Yale Program on Financial Stability at ypfs@yale.edu .
